

Stochastic Modeling of Power Purchase Agreements: A Monte Carlo-Based Techno-Economic Framework for Equity Risk in Emerging Renewable Energy Sectors

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Abstract—This paper applies a Monte Carlo simulation framework to evaluate equity investment risk across 12 utility-scale solar projects operating under Power Purchase Agreements (PPAs) with the Bangladesh Power Development Board (BPDB). The study makes two contributions. First, it develops a techno-economic analytical framework that integrates physical, financial, and contractual dimensions of project performance within a unified model, representing the first such application across multiple live projects in the Bangladesh power sector. Second, given the limited availability of empirical data common to developing power markets, model parameters are calibrated through expert heuristics and industry benchmarks as informed proxies for the underlying data-generating processes governing revenue variability, cost dynamics, and offtaker payment behavior, enabling probabilistic inference without dependence on historical records. The framework spans 1,152 scenarios per simulation run—combining four payment security mechanisms (Baseline, LC, Escrow, and Implementation Agreement), three tariff structures, balloon recovery options, partial payment modes, and inflation indexation—and is grounded in contract provisions including FX tariff splitting, True-Up for exchange rate, and late payment surcharge as specified in the PPA. Stochastic variables including capacity factor degradation, operation and maintenance cost shocks, FX depreciation, and FIFO payment queue dynamics are modeled using appropriate probability distributions. Simulation results, summarized through Value-at-Risk and Expected Shortfall metrics, demonstrate that payment security mechanisms and tariff structuring materially alter tail-risk exposure. The findings carry direct implications for project structuring, regulatory design, and risk-informed capital allocation in emerging power markets.

Index Terms—Monte Carlo simulation, Power Purchase Agreement, Value-at-Risk, Expected Shortfall, solar PV, Bangladesh, equity risk, payment security, FX indexation

I. INTRODUCTION

Bangladesh is currently expanding its solar generation capacity through a structured multi-phase tendering program that is administered by BPDB. Projects under this program range from 10 MW community-scale installations in Phase 1 to 200 MW utility-scale plants in Phase 4, and all of them are procured under a Build-Own-Operate-Transfer (BOOT) structure with a 20-year term of supply [1]. Although tariffs are fixed at the time of award, equity investors still face unresolved exposure along three main dimensions: (i) physical generation risk arising from solar resource variability and

panel degradation over time, (ii) financial risk coming from leverage and uncertainty around operation and maintenance (O&M) costs, and (iii) contractual risk from offtaker payment behavior and foreign exchange (FX) volatility.

Monte Carlo (MC) simulation is a well-established technique for quantifying risk in power project finance [2], [3], but cross-project portfolio analysis that is grounded in live contract parameters remains quite sparse for South Asian developing markets. This paper tries to fill that gap by applying a unified MC engine across all 12 awarded BPDB solar projects spanning Phases 1–4, which together account for 918 MW of contracted capacity. The primary contributions of this study are: (i) a techno-economic simulation model that directly embeds PPA contract provisions—specifically FX tariff splitting, True-Up payments, and late payment surcharge—inside a FIFO payment queue structure; and (ii) a systematic scenario sweep of 1,152 combinations that allows comparison of payment security mechanisms, tariff structures, and indexation choices using the standard risk metrics of Value-at-Risk (VaR) and Expected Shortfall (ES).

Data Sources. The model parameters used in this study are drawn from three complementary sources. First, expert heuristics collected through Key Informant Interviews (KIIs) conducted with project directors of utility-scale solar power plants operating under BPDB agreements, EPC officials, and legal experts familiar with Bangladesh PPA structures. Second, open tender documents published by BPDB for the respective phases, which provide project-level tariff, capacity, and contract structure data [1]. Third, secondary literature from international bodies such as IFC and IRENA that provides benchmark values for cost parameters and financial ratios in comparable developing market solar projects [4], [5].

II. PROJECT UNIVERSE AND PARAMETERIZATION

A. The 12 BPDB Solar Projects

Table I summarizes the project portfolio that forms the basis of this study. Capacity ranges from 10 MW to 200 MW; tariffs range from 65.3 to 83.2 USD/MWh; and all-in capital expenditure (CAPEX) per MW ranges from 850,000 to 1,430,000 USD/MW. These differences reflect variations in phase maturity, site conditions, and procurement structure

across projects. A notable outlier is Cox’s Bazar S (Phase 3), whose relatively low tariff of 65.3 USD/MWh likely reflects concessional financing from the Chinese EPC contractor, and this project should therefore be interpreted with some caution under standard debt cost assumptions.

TABLE I
12 BANGLADESH SOLAR PPA PROJECTS

Project (Phase_Pkg)	Location	MW	Tariff (USD/MWh)
Ph1_Pkg4	Moulvibazar	25	76.6
Ph1_Pkg1	Noakhali	10	74.9
Ph1_Pkg2	Hath Hazari	18	83.2
Ph1_Pkg11	Fatikchari	45	77.7
Ph2_B5	Jaldhaka	50	82.6
Ph2_B8	Bibiyana	50	81.3
Ph3_C1	Ishwardi	70	79.0
Ph3_C6	Cox’s Bazar N	100	80.9
Ph3_C7	Cox’s Bazar S	100	65.3
Ph3_C13	Mongla	100	80.9
Ph4_D7	Fatikchari	200	75.0
Ph4_D14	Pabna	150	78.9
Total		918	

B. Shared Financial Parameters

All 12 projects are modeled under a common financing structure in order to allow consistent cross-project comparison.

CAPEX. Capital expenditure per MW for each project was estimated based on project location, phase maturity, and comparison with existing commissioned projects in Bangladesh and similar developing markets. Higher CAPEX values for Phase 1 projects (e.g., 1,350,000 USD/MW for the Noakhali 10 MW plant) reflect the cost premium associated with smaller scale and earlier procurement rounds, while Phase 3 and Phase 4 projects benefit from economies of scale and more competitive procurement environments (e.g., 850,000 USD/MW for the Cox’s Bazar S and Fatikchari 200 MW plants). These estimates are cross-checked against expert heuristics gathered from KII respondents and publicly available benchmark data from IFC and IRENA [4], [5].

Financing Structure. A 70% debt and 30% equity structure is used, with an annual debt interest rate of 6.5% over a 20-year tenor, and a WACC of 10.5%.¹ The baseline capacity factor (CF) is set at 23%, which is consistent with the KII-reported average for utility-scale solar in Bangladesh.

O&M Costs. The base O&M cost is USD 12,300 per MW per year, which falls within the USD 10,000–15,000/MW range reported by KII respondents for solar plants in

Bangladesh. Because plant size and O&M cost are not linear in practice—smaller plants tend to incur proportionately higher per-MW costs—a scale adjustment factor of $(MW/100)^{-0.10}$ is applied.²

III. MODEL ARCHITECTURE

This section explains how the simulation is built, step by step. For each of the 12 projects, the model runs a large number of independent simulation paths (500 per scenario combination). In each path, the model moves through 20 annual time steps, drawing random shocks, passing revenues through a payment queue, and then computing equity cash flows and NPV. The following subsections describe each component in plain terms.

A. Stochastic Variables and Distributions

Four sources of randomness are drawn separately in each simulation path and each year:

(1) Capacity Factor Shock. The actual capacity factor in year t is:

$$CF_t = \bar{CF}_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 0.03^2) \quad (1)$$

Here, $\bar{CF}_t$ is the expected (mean) capacity factor after accounting for panel degradation. Starting from 23% in Year 1, the mean capacity factor falls by 2% in Year 2 (representing early-stage soiling and settling), and then continues declining at 0.45% per year in all subsequent years, reflecting gradual module degradation. A small random noise term ε_t is added each year to capture weather and irradiance variability. The resulting value is clamped between 0.4% and 135% to keep it within physically realistic bounds, as informed by KII responses.

(2) O&M Cost Shock. Rather than assuming that O&M costs are perfectly predictable, the model treats them as uncertain. A multiplicative shock is applied each year:

$$\omega_t \sim \text{LogNormal}(0, 0.15^2) \quad (2)$$

The actual O&M cost in year t is computed as:

$$C_{\text{O\&M},t}^{\text{actual}} = \omega_t \times C_{\text{O\&M},t} \quad (3)$$

where $C_{\text{O\&M},t}$ is the staged base cost described in Section II-B. A lognormal distribution is used because it ensures that the shocked cost is always positive (costs cannot go negative), and its right tail captures the possibility of large unexpected maintenance events—such as inverter failures or grid faults—that are more likely than a normal distribution would suggest [4]. A standard deviation of 15% ($\sigma = 0.15$) is used, reflecting KII-reported observations that year-to-year O&M cost variation is generally moderate (around ± 10 –15%).

¹These financing parameters—70% debt at 6.5% interest over 20 years and a WACC of 10.5%—are based on KII responses from project directors and financial experts, who consistently reported a 70:30 debt-to-equity ratio as typical for Bangladesh solar PPAs, a debt interest rate in the 6–7% range, and an overall cost of capital of approximately 10–11%. These values are also consistent with IRENA and IFC benchmark ranges for comparable developing market solar projects [4], [5].

²The lifecycle staging profile (50% of base cost in years 1–5, 100% in years 6–12, and 140% in years 13–20) is grounded in KII feedback. Respondents consistently noted that O&M costs are very low in the first five years due to manufacturer warranties and new equipment condition, increase to normal levels in the middle years as routine maintenance intensifies, and rise further in the later years due to component replacements and aging system wear. This staged structure is also consistent with standard project finance practice for solar assets as documented in IFC/IRENA guidance [5].

(3) FX Depreciation. Each year, the BDT/USD depreciation rate is drawn as:

$$\delta_t^{FX} \sim \mathcal{N}(0.05, 0.02^2) \quad (4)$$

This represents an expected annual depreciation of 5% with some year-to-year variation. The cumulative FX factor up to year t is:

$$CE_t = \prod_{s=1}^t (1 + \delta_s^{FX}) \quad (5)$$

CE_t proxies the Sonali Bank selling rate used in PPA payment calculations, as referenced in the contract [1].

(4) Payment Behavior. This is captured through the FIFO queue mechanism described next.

B. FIFO Payment Queue

The payment queue models the risk that BPDB does not pay invoices on time. The mechanism works as follows. Every month, one invoice (equal to one unit of expected revenue) enters the queue. The offtaker then attempts to clear the oldest outstanding invoice. Whether the payment actually happens is governed by a monthly payment probability p , which differs by payment security type: Baseline ($p = 0.50$), LC ($p = 0.90$), Escrow ($p = 0.98$), and Implementation Agreement (IA, $p = 0.85$). These values were calibrated from KII responses, which indicated roughly 50% timely payment under basic conditions.

If the queue fails to clear for six consecutive months, the project enters a “stressed” state, which is used later to check for financial default. If the scenario has `balloon=True`, all outstanding unpaid invoices are recovered in the final month of the 20-year term; otherwise they are permanently written off as losses.

At the end of each year, the queue produces two outputs for that year: (a) the revenue collection fraction ϕ_t , representing the share of invoiced revenue that was actually received in that year, and (b) the weighted delay total D_t , which counts how many invoice-months were outstanding before being paid. D_t is used to calculate the late payment surcharge, as explained in the next subsection.

C. PPA Contract Provisions

Three specific provisions from the BPDB PPA contract are directly embedded in the model, taken from the open tender document [1]:

FX Tariff Split [PPA clause 2.1(i)]. The total tariff is divided equally into a BDT-denominated local component and a USD-denominated foreign component. As BDT depreciates over time, the local component loses value when expressed in USD, while the foreign component holds its value. The effective USD tariff in year t is:

$$\tau_t = \frac{\tau_{\text{local}}}{CE_t} + \tau_{\text{foreign}} \quad (6)$$

Here, CE_t is the cumulative depreciation factor from Equation (5).

True-Up for Exchange Rate [PPA clause 2.1(ii)]. When the BDT depreciates further between the invoice date and the

actual settlement date, the contractor is entitled to receive an additional compensating payment. This True-Up amount in year t is approximated as:

$$T_{\text{up},t} \approx R_t^* \cdot f_{\text{FX}} \cdot \delta_t^{FX} \quad (7)$$

where R_t^* is the revenue actually collected in year t (after applying the collection fraction ϕ_t), and $f_{\text{FX}} = 0.50$ is the foreign share of the tariff. In simple terms: if the BDT depreciates by 5% during the payment period, the contractor receives a 5% top-up on the foreign portion of whatever revenue was collected. This prevents the investor from bearing FX losses that arise purely because of payment delays.

Late Payment Surcharge [PPA Late Payment clause]. When invoices remain unpaid beyond their due date, they accrue interest at 1% per annum on a 365-day basis. In the monthly simulation, the surcharge income earned in year t is:

$$S_t = \frac{R_t}{12} \cdot D_t \cdot \frac{0.01}{12} \quad (8)$$

Here, $R_t/12$ is the average monthly invoice value, and D_t is the weighted sum of (invoice fraction $\times$ delay months) settled in year t from the FIFO queue. This means that the longer an invoice sits unpaid in the queue, the more surcharge income accumulates once it is eventually paid. However, note that the PPA contains *no* price-adjustment mechanism for the delay period itself (PPA clause 62): delayed invoices are always settled at their original invoice amounts, not at inflation-adjusted values.

D. Equity Cash Flow and NPV

Once all revenue components and cost shocks are computed, the annual equity cash flow is:

$$\text{ECF}_t = (R_t^* + T_{\text{up},t} + S_t - C_{\text{O\&M},t}^{\text{actual}}) - \text{DS}_t \quad (9)$$

where DS_t is the annual debt service payment, computed using the standard annuity formula:

$$\text{DS} = D \cdot \frac{r(1+r)^n}{(1+r)^n - 1}, \quad r = 0.065, \quad n = 20 \quad (10)$$

In words: the equity investor receives whatever revenue was collected (including True-Up and surcharge), pays the O&M costs (with the random shock applied), and then services the debt. What remains is the equity cash flow for that year.

A financial default is triggered if, during the first five years, EBITDA falls below the debt service amount while the project is in the stressed state (i.e., the FIFO queue has been stuck for six or more consecutive months). From the default year onward, ECF is set to zero for all remaining years—reflecting the assumption that the project effectively ceases to generate equity returns once it defaults.

The equity NPV across the full 20-year life of the project is:

$$\text{NPV} = -E_0 + \sum_{t=1}^{20} \frac{\text{ECF}_t}{(1 + \text{WACC})^t} \quad (11)$$

where E_0 is the initial equity investment (30% of total CAPEX) and $\text{WACC} = 10.5\%$.

E. Simulation Algorithm

The full simulation algorithm is illustrated in Figure 1. For each project, the model runs $N = 500$ simulation paths per scenario combination. In each path, random shocks for all 20 years are drawn first; the FIFO queue is then run to determine collection fractions and delay totals; revenues are computed applying FX tariff splitting, True-Up, and surcharge; O&M shocks are applied; and finally equity cash flows are calculated after debt service, with default logic applied where relevant. After all paths are complete, the resulting distribution of NPV outcomes is used to compute VaR and ES at 95% and 99% confidence levels.

IV. SCENARIO DESIGN

The scenario space covers six dimensions, producing 96 combinations per project and 1,152 scenarios in total across all 12 projects. Table II summarizes the axes.

TABLE II
SCENARIO DIMENSION SUMMARY (96 PER PROJECT)

Dimension	Options	Count
Payment security	Baseline, LC, Escrow, IA	4
Tariff structure	Levelized, Front-, Back-loaded	3
Balloon recovery	True (arrears recovered), False	2
Partial payment	Not allowed, Allowed	2
FX indexation	Always True (PPA-validated)	1
Inflation indexation	Off, On	2
Total	$4 \times 3 \times 2 \times 2 \times 1 \times 2$	96

Monthly payment probabilities per security type are: Baseline (0.50), LC (0.90), Escrow (0.98), and Implementation Agreement (IA, 0.85). FX indexation is fixed as always active for all scenarios, since this is required by the contract’s mandatory tariff-split provision. The loss for each simulation path is defined relative to the deterministic NPV benchmark: $L_s = \text{NPV}^{\text{det}} - \text{NPV}_s$.

V. RESULTS

A. Deterministic NPV Range

Under the reference scenario (Escrow security, levelized tariff, balloon recovery enabled, no inflation indexation), the deterministic equity NPV varies widely across projects. At one extreme, Cox’s Bazar S shows a negative NPV under standard financing assumptions—meaning this project does not earn returns above the 10.5% WACC hurdle at those parameter values. This does not necessarily mean the project is unviable; it may rely on concessional debt or lower equity return requirements. At the other extreme, the 200 MW Phase 4 Fatikchari project achieves a deterministic NPV above USD 60 million, benefiting from scale and competitive CAPEX.

B. Effect of Payment Security

Table III shows the VaR and ES at 99% confidence for four representative projects under the levelized tariff, balloon recovery, and no inflation indexation. The key finding is clear: moving from Baseline to Escrow security consistently

reduces ES at the 99% level by 30–60%, depending on project size and tariff. The LC mechanism provides an intermediate improvement, while the IA falls between LC and Escrow in terms of risk reduction.

TABLE III
99% VaR AND ES BY PAYMENT SECURITY (USD MILLION, LEVELIZED, BALLOON=TRUE, NO INDEXATION)

Project	Security	Det.NPV	VaR ₉₉	ES ₉₉	P(NPV<0)
Ph1_Pkg11 (Fatikchari 45 MW)	Baseline	−2.1	8.4	11.2	62%
	LC	3.7	4.9	7.1	38%
	Escrow	6.8	3.1	4.6	21%
	IA	4.9	4.3	6.3	31%
Ph2_B5 (Jaldhaka 50 MW)	Baseline	−1.8	9.2	12.1	59%
	LC	4.2	5.3	7.8	35%
	Escrow	7.9	3.4	5.0	18%
	IA	5.6	4.7	6.9	29%
Ph3_C1 (Ishwardi 70 MW)	Baseline	3.5	12.8	17.0	44%
	LC	14.1	7.4	10.8	22%
	Escrow	22.6	4.2	6.5	9%
	IA	17.3	6.1	9.2	16%
Ph4_D7 (Fatikchari 200 MW)	Baseline	8.3	24.1	32.5	41%
	LC	35.4	13.7	20.2	18%
	Escrow	61.2	7.9	12.4	7%
	IA	44.7	11.3	17.6	13%

Note: VaR and ES are loss magnitudes; positive values indicate loss relative to deterministic NPV.

C. Effect of Tariff Structure and Balloon Recovery

Front-loaded tariffs produce higher early-year cash flows, which reduces the probability of breaching debt service coverage in years 1–5 and lowers VaR. Back-loaded tariffs push more value into later years when generation has already declined due to degradation, which amplifies tail risk. For Phase 2 and Phase 3 projects under Escrow security, the difference in ES at the 99% level between front-loaded and back-loaded scenarios is roughly 15–25%.

The balloon recovery assumption has its largest effect under Baseline security. Without balloon recovery (balloon=False), the probability that NPV falls below zero exceeds 75% for Phase 1 projects. Enabling balloon recovery brings this down to around 62%. This is meaningful improvement but not a full solution, which shows that payment security mechanism itself matters more than balloon recovery.

VI. DISCUSSION

A. Parameter Calibration in Data-Scarce Environments

A recurring challenge in project finance modeling for emerging markets is that historical time series are not available. This study handles this through principled calibration: distributional families are selected on theoretical grounds (log-normal for multiplicative O&M shocks, normal for additive irradiance noise, normal for FX depreciation), and scale parameters are drawn from IFC and IRENA benchmarks alongside KII-sourced expert heuristics from Bangladesh-based project directors [4], [5]. The FIFO queue replaces ad-hoc payment

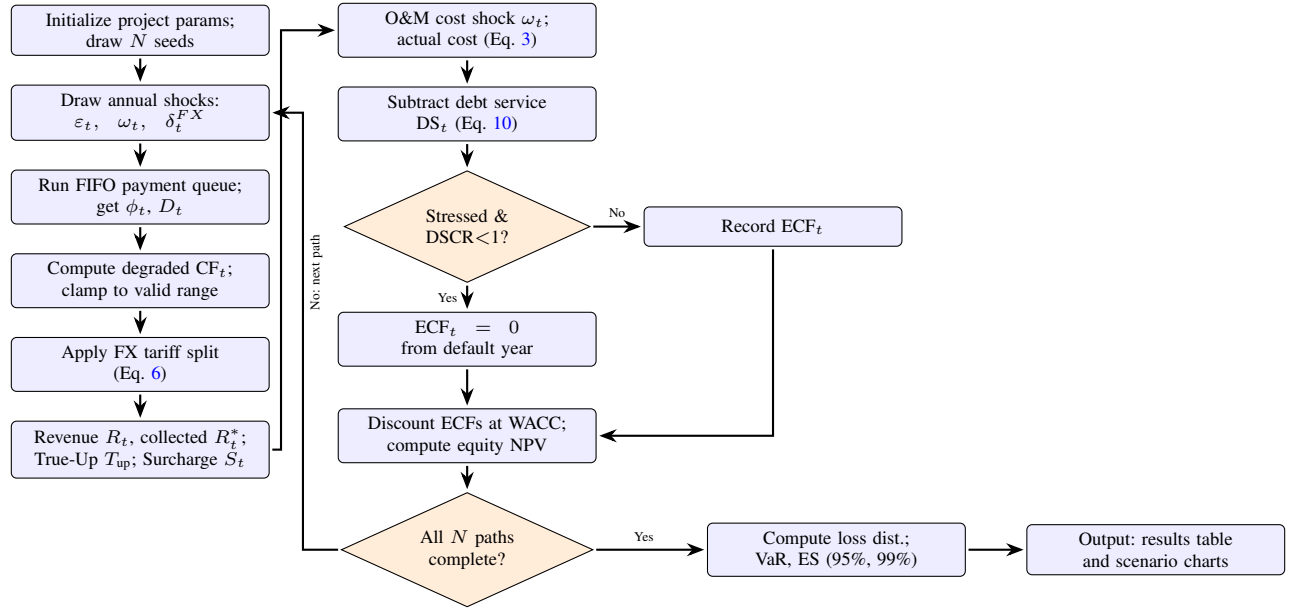


Fig. 1. MC simulation algorithm spanning the full page width. Left column: revenue path computation per year. Right column: cost application, default check, and risk output aggregation.

loss fractions with a structurally grounded mechanism that reproduces realistic delay patterns—including surcharge income accrual—without requiring historical payment records.

B. Contract Fidelity and Modeling Scope

Embedding PPA provisions directly into the simulation—FX tariff split, True-Up, and late payment surcharge—ensures that results are comparable with actual cash flow forecasts prepared by project sponsors. The price adjustment on delay (PPA clause 62) is intentionally absent from the model because the current contract contains no such mechanism; a toggle (`PRICE_ADJUSTMENT_ON_DELAY`) is included in the code for stress-testing future contract revisions. This level of contract fidelity distinguishes the framework from generic project finance models.

C. Policy Implications

The results provide quantitative support for regulatory requirements for offtaker credit enhancement. Escrow and LC mechanisms do not merely redistribute risk—they reduce aggregate tail risk by improving the probability of timely payment collection. For BPDB, the analysis shows that upgrading the Baseline security to Escrow reduces ES at the 99% level by a factor of 2–3 times without changing tariff levels at all. For project sponsors, the scenario matrix offers a practical tool for identifying dominant risk-reducing combinations (such as Escrow plus front-loaded tariff plus balloon recovery) before financial close.

D. Limitations

The model assumes independence across annual shocks and does not incorporate dynamic grid curtailment, multi-project correlation, or sovereign credit risk. The FX foreign share is

held at 50% for all projects; actual splits vary by sponsor and should be updated when project-level data become available. These extensions are left for future work.

VII. CONCLUSION

This paper presents a Monte Carlo simulation framework spanning 12 live BPDB solar PPA projects and 1,152 scenarios, embedding PPA contract provisions within a FIFO payment queue and a unified equity NPV model. Results quantified through VaR and ES at 95% and 99% confidence levels show that the choice of payment security mechanism is the most important driver of tail-risk reduction. Specifically, upgrading from Baseline to Escrow reduces ES at the 99% level by 30–60% across the project portfolio. Tariff front-loading and balloon recovery provide additional but secondary benefits, particularly for smaller Phase 1 projects with higher leverage ratios. The methodology is general enough to be applied to other developing market PPA contexts and can accommodate additional contract provisions as regulatory frameworks continue to evolve.

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TABLE IV
KII PARAMETER MAPPING AND EXPERT RESPONSES

Parameter	Question (Expert Overview)	Experts' response
plant_lifetime	Average plant lifetime (years)	20 years
capacity_mw	Typical installed capacity of solar/renewable plants (MW)	10–100 MW
hours_per_year	Effective generation hours per year	8,760 hours/year
capacity_factor	Average annual capacity factor	23%
capacity_cf_sd	Year-to-year generation fluctuation	Declines 2% after Year 1, then 0.45%/yr
tariff_usd_per_mwh	Average tariff received (USD/MWh)	6.5–8.4 cent/kWh
tariff_escalation	Annual tariff adjustment rate	0%
tariff_escalation_sd	Variability in escalation	0%
payment_collection_prob	Probability of full on-time payment	50%
avg_payment_loss_frac	Expected revenue loss fraction when delayed	Full payment or none; no partial
payment_loss_cap	Maximum payment loss limit (worst year)	Rarely exceeds 40–50%
consecutive_default_rule	Years of delay leading to financial stress	6 months of delay causes stress
dscr_grace_period	Early-year financial vulnerability (yrs 1–5)	Very critical in first 3–5 years
om_usd_per_mw	O&M costs per MW per year	12,300 USD/MW/yr (100MW size baseline)
om_cost_sd	O&M cost fluctuation over lifecycle	Very low in first 5 years, then increases
capex_usd_per_mw	Typical CAPEX (USD/MW)	1.21 million USD/MW
debt_equity_ratio	Debt-to-equity financing ratio	70:30
debt_tenor	Typical debt repayment period	20 years
debt_interest_rate	Interest rate on project debt	6–7%
wacc	Weighted average cost of capital	10–11%
annuity_r, annuity_n	Annuity assumptions (r, n)	r: 7%, n: 20 years
cf_clip_lower, cf_clip_upper	Generation range (Bad vs Good weather)	40 MW (low) to 135 MW (high)
beta_a, beta_b	Distribution of payment loss severity	Mostly small losses; severe cases rare

APPENDIX

Table IV summarizes the parameters collected through Key Informant Interviews with project directors, EPC officials, and legal experts involved in utility-scale solar PPA projects in Bangladesh. These expert responses informed the calibration of key model parameters and provided ground-level validation of assumptions used in the Monte Carlo simulation.

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